CIS 613: MC/DC Testing

Modified Condition Decision Coverage Testing

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Definitions

Boolean Expression: Evaluates to one of two possible outcomes (e.g., true and false)

$$((A \ \wedge B) \ \vee (4 < 3)) \wedge \neg A$$

2

Definitions

Condition: Operand or argument to a Boolean operator.

$$((A \land B) \lor (4 < 3)) \land \neg A$$

3

Definitions

Coupled Condition: One condition changes another.

$$((A \land B) \lor (4 < 3)) \land \neg A$$

4

Definitions

Masking Conditions: One condition masks or hides the impact of another.

$$A \wedge false$$

$$A \vee true$$

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MCDC Coverage

Modified Condition Decision Coverage (MCDC)

- Every program entry and exit point is invoked at least once.
- All possible outcome of every control statement are taken.
- Each condition in a control statement takes every possible outcome.
- Every (non-constant) condition in a control statement has been shown to *independently* affect the outcomes of the control statement.

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MCDC Coverage

How does MCDC Coverage compare to:

- Line Coverage
- Branch Coverage
- Condition Coverage
- Branch/Condition Coverage
- Multiple Condition Coverage

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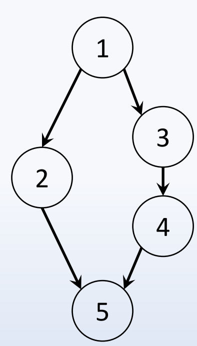
Independent Effect

Unique-Cause MCDC: Condition causes decision change

```
1)  if(a && (b || c)) {
2)      y = 1;
3)  } else {
4)      y = 2;
5)  }
```



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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***a*** to change the result?

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Independent Effect

How do we toggle ***a*** to change the result?
If we choose R_1, select one from 2 – 8:

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***a*** to change the result?

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***b*** to change the result?

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Independent Effect

How do we toggle ***b*** to change the result?
If we choose R_2, select one from 1 – 8:

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***b*** to change the result?

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Independent Effect

How do we toggle ***c*** to change the result?
If we choose R_3, select one from 1 – 8:

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***c*** to change the result?

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Independent Effect

Can we always cause an independent effect?

Are these the same?

```
1)  if(a && (b || c)) {
2)      y = 1;
3)  } else {
4)      y = 2;
5)  }
```

```
1)  if((a && b) || (a && c)) {
2)      y = 1;
3)  } else {
4)      y = 2;
5)  }
```

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
$\boldsymbol{a} \wedge (\boldsymbol{b} \vee \boldsymbol{c})$	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**F**
$(\boldsymbol{a} \wedge \boldsymbol{b}) \vee (\boldsymbol{a} \wedge \boldsymbol{c})$	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**F**

Yup, they are the same!

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a_1	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
a_2	T	T	T	T	F	F	F	F
$(\boldsymbol{a_1} \vee \boldsymbol{b}) \wedge (\boldsymbol{a_2} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle $\boldsymbol{a_1}$ to change the result?

But that is ridiculous! We know they are the same variable!

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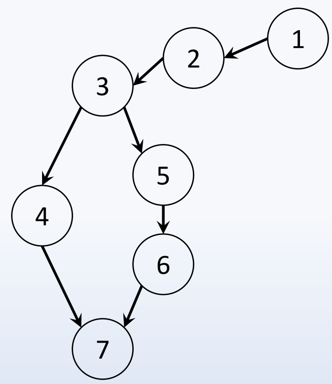
Independent Effect

Unique-Cause MCDC: Condition causes decision change

1) d = a;
2) …
3) if((a && b) || (**d** && c)) {
4)   y = 1;
5) } else {
6)   y = 2;
7) }



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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}	R_{11}	R_{12}	R_{13}	R_{14}	R_{15}	R_{16}
a	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
d	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
$(\boldsymbol{a} \vee \boldsymbol{b}) \wedge (\boldsymbol{d} \vee \boldsymbol{c})$	T	T	T	F	F	F	F	F	T	T	F	F	F	F	F	F
y=1	X	X	X						X	X						
y=2				X	X	X	X	X			X	X	X	X	X	X

How do we toggle ***a*** to change the result?

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}	R_{11}	R_{12}	R_{13}	R_{14}	R_{15}	R_{16}
a	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
d	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
$(a \vee b) \wedge (d \vee c)$	T	T	T	F	F	F	F	F	T	T	F	F	F	F	F	F
y=1	X	X	X						X	X						
y=2				X	X	X	X	X			X	X	X	X	X	X

Slight problem…

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Independent Effect

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
a	T	T	T	T	F	F	F	F
b	T	T	F	F	T	T	F	F
c	T	F	T	F	T	F	T	F
d	T	T	T	T	F	F	F	F
$(a \vee b) \wedge (d \vee c)$	T	T	T	F	F	F	F	F
y=1	X	X	X					
y=2				X	X	X	X	X

How do we toggle ***a*** to change the result?

We can't! We have to *mask* it!

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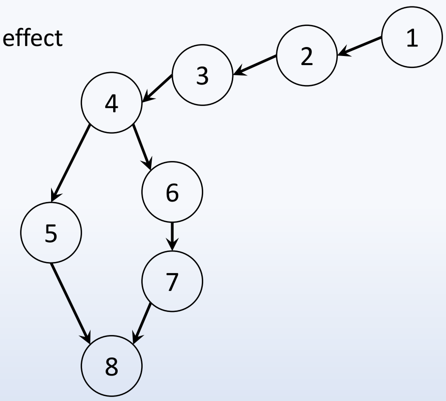
Masking Coupled Conditions

Unique-Cause MCDC: Condition causes decision change

- Add a variable to *mask* a condition
- Effectively, the *masked* variable has no effect

```
1) e = setByTest();
2) d = a;
3) …
4) if((a && b) || ((d || e) && c)) {
5)     y = 1;
6) } else {
7)     y = 2;
8) }
```



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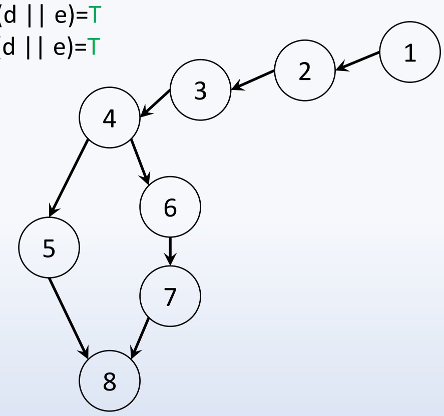
Masking Coupled Conditions

Unique-Cause MCDC: Condition causes decision change

Test 1: a = T, b = T, c = T, d = T, e = T → (d || e)=T
Test 2: a = F, b = T, c = T, d = F, e = T → (d || e)=T

```
1) e = setByTest();
2) d = a;
3) …
4) if((a && b) || ((d || e) && c)) {
5)     y = 1;
6) } else {
7)     y = 2;
8) }
```



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